Vector Calculus and Complex Analysis Classwork (2025/5/15)

Name:

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Let $m > n$. Use Cauchy's integral formula for derivatives to compute $\int_{|z-1|=r} \frac{z^m}{(z-1)^{n+1}} dz$. Hence show that $\binom{m}{n} \leq \frac{m^m}{n^n (m-n)^{m-n}}$.

Solution

$$\int_{|z-1|=r} \frac{z^m}{(z-1)^{n+1}}\,dz = \frac{2\pi i}{n!}\frac{d^n}{dz^n} z^m\Big|_{z=1} = \frac{2\pi i}{n!} m(m-1)\cdots(m-n+1) = 2\pi i \binom{m}{n}$$

$$|z-1| = r \quad \Rightarrow \quad |z| = |z-1+1| \le |z-1|+1 = r+1$$

$$\left|2\pi i \binom{m}{n}\right| = \left|\int_{|z-1|=r} \frac{z^m}{(z-1)^{n+1}}\,dz\right| \le 2\pi r \frac{(r+1)^m}{r^{n+1}}$$

$$\binom{m}{n} \le \frac{(r+1)^m}{r^n} \quad \text{for all } r > 0.$$

Let $f(r) = \dfrac{(r+1)^m}{r^n}$.

$$f'(r) = \frac{r^n m(r+1)^{m-1} - (r+1)^m n r^{n-1}}{r^{2n}} = (r+1)^{m-1}\frac{mr-(r+1)n}{r^{n+1}}$$

r	$< \dfrac{n}{m-n}$	$= \dfrac{n}{m-n}$	$> \dfrac{n}{m-n}$
$f'(r)$	< 0	$= 0$	> 0

$f(r)$ is minimum at $r = \dfrac{n}{m-n}$

$$\binom{m}{n} \le \frac{\left(\dfrac{m}{m-n}\right)^m}{\left(\dfrac{n}{m-n}\right)^n} = \frac{m^m}{n^n (m-n)^{m-n}}.$$